

Team Project: Astro Track

Advanced Database Concepts

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Problem Domain Description:

This database system is proposed to help astronomical research by storing, analyzing data and managing. Indeed, it provides 10 tables, consisting of 7 entity tables and 3 bridge tables. According to Entity Tables, there are celestial objects, events, affiliations, research papers, researchers, telescopes, and missions. According to Bridge Tables, there are observations, mission observations and habitable planets.

1. Celestial Objects:

- This table is proposed to track various types of celestial objects.
- No foreign keys.

2. Events:

- This table is proposed to capture significant astronomical occurrences.
- No foreign keys.

3. Affiliations:

- This table is proposed to list institutions which conduct space research.
- No foreign keys.

4. Researchers:

- This table is proposed to store researchers' details.
- It has a foreign key that references the Affiliations table.

5. Research Papers:

- This table is proposed to catalog articles published by researchers.
- It has a foreign key that references the Researchers table.

6. Telescopes:

- This table is proposed to manage observation data collected by telescopes.
- No foreign keys.

7. Missions:

- This table is proposed to manage records of mission in space.
- It has a foreign key that references the Researchers and Affiliations table.

8. Observations:

- This table is proposed to collect the specific observation in space based on Celestial Objects, Researchers, and Telescopes tables.
- It will provide many-to-many relational structures.

- It has foreign keys that reference the Celestial Objects, Researchers, and Telescopes table.

9. Mission Observations:

- This table is proposed to record detailed information about each observation's role within a mission based on Missions and Observations tables.
- It will provide many-to-many relational structures.
- It has foreign keys that reference the Missions and Observations tables.

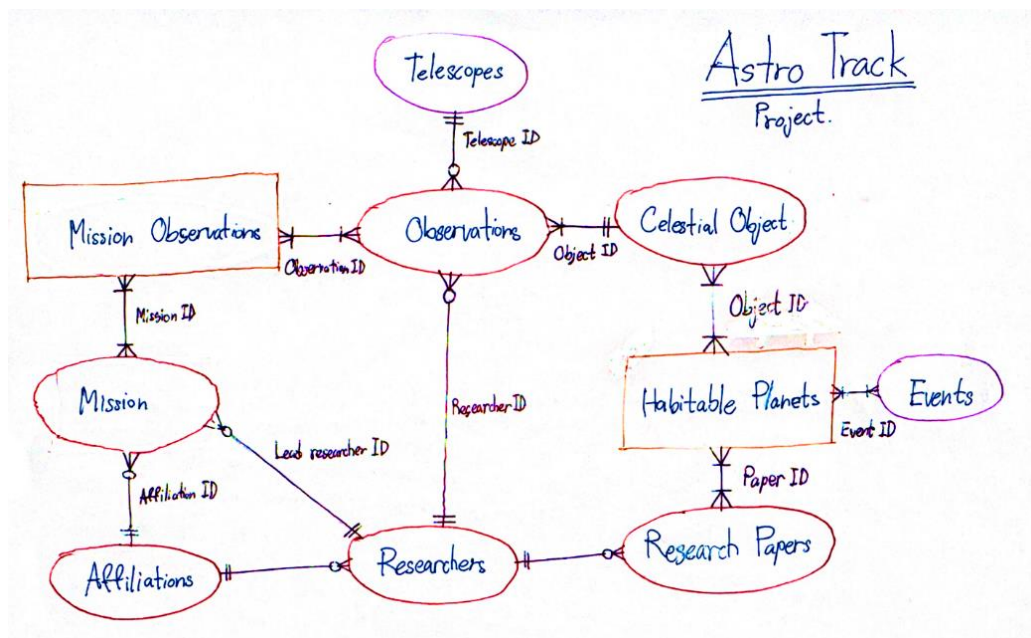
10. Habitable Planets:

- This table is proposed to evaluate the habitability potential of planets or exoplanets based on Celestial Objects table, Events, and Research Papers tables.
- It will provide many-to-many relational structures.
- It has foreign keys that reference the Celestial Objects, Events, Research Papers table.

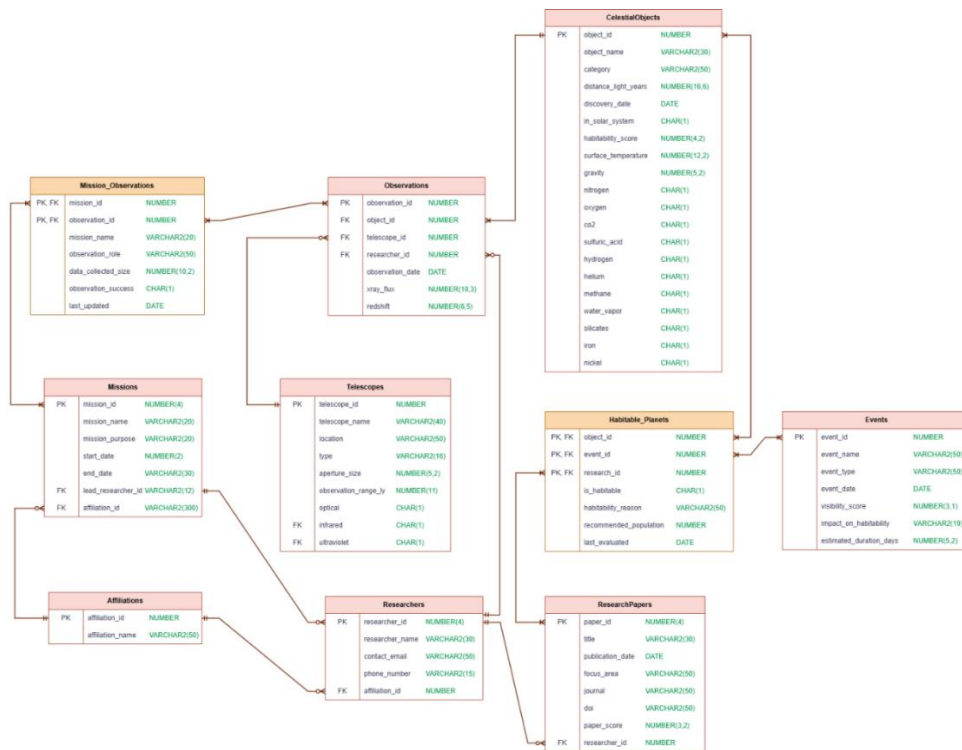
ER Diagram:

The Astro track project has started with Mind Mapping which mentions only the name of table in order to design the connection between the entity and bridge tables. Then we used Microsoft Visio to define details of columns in each table. Once the table structures are confirmed within our team, we start to create all tables on Oracle SQL Developer and provide as rational models diagram.

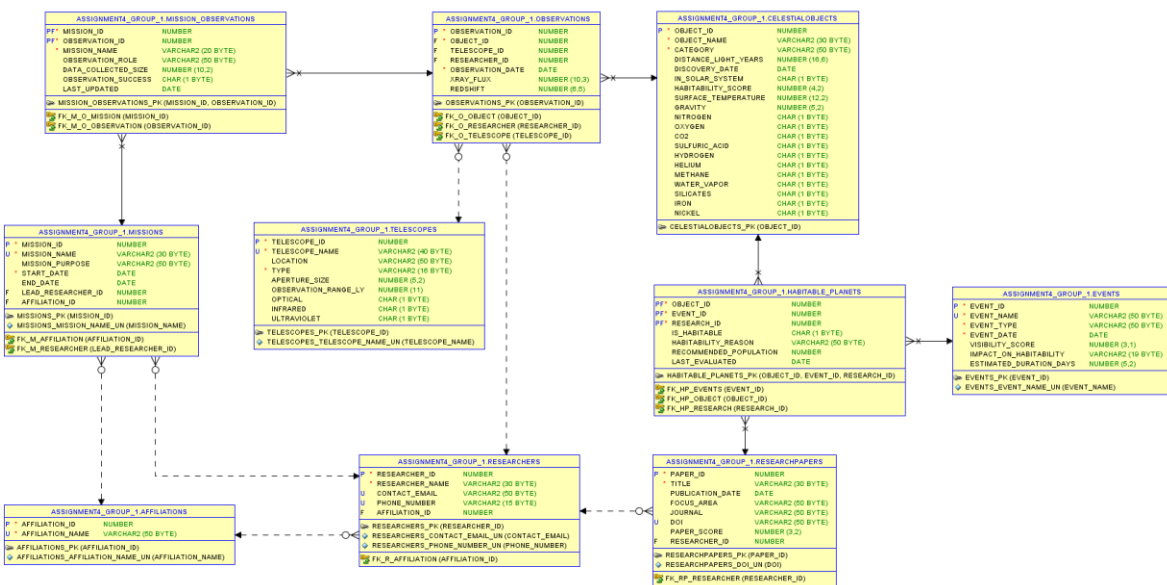
1. Mind Mapping – ER Diagram:



2. VISIO – ER Diagram:

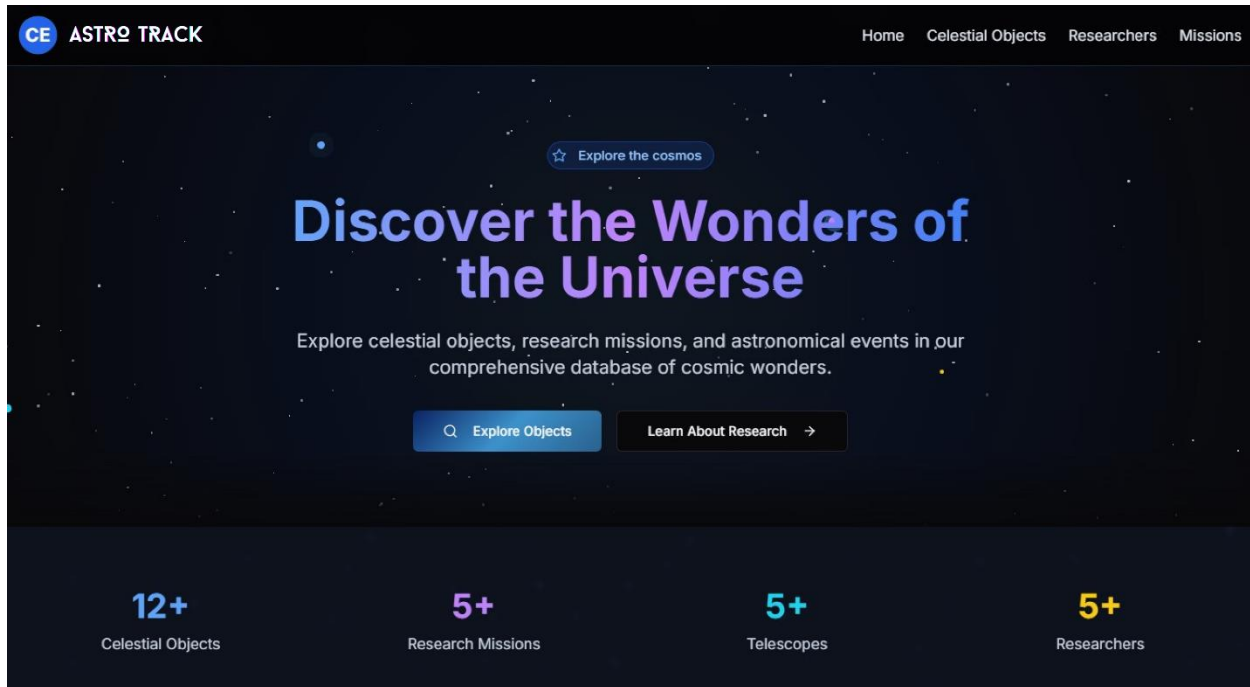


3. Oracle SQL Developer – ER Diagram/Rational Model:



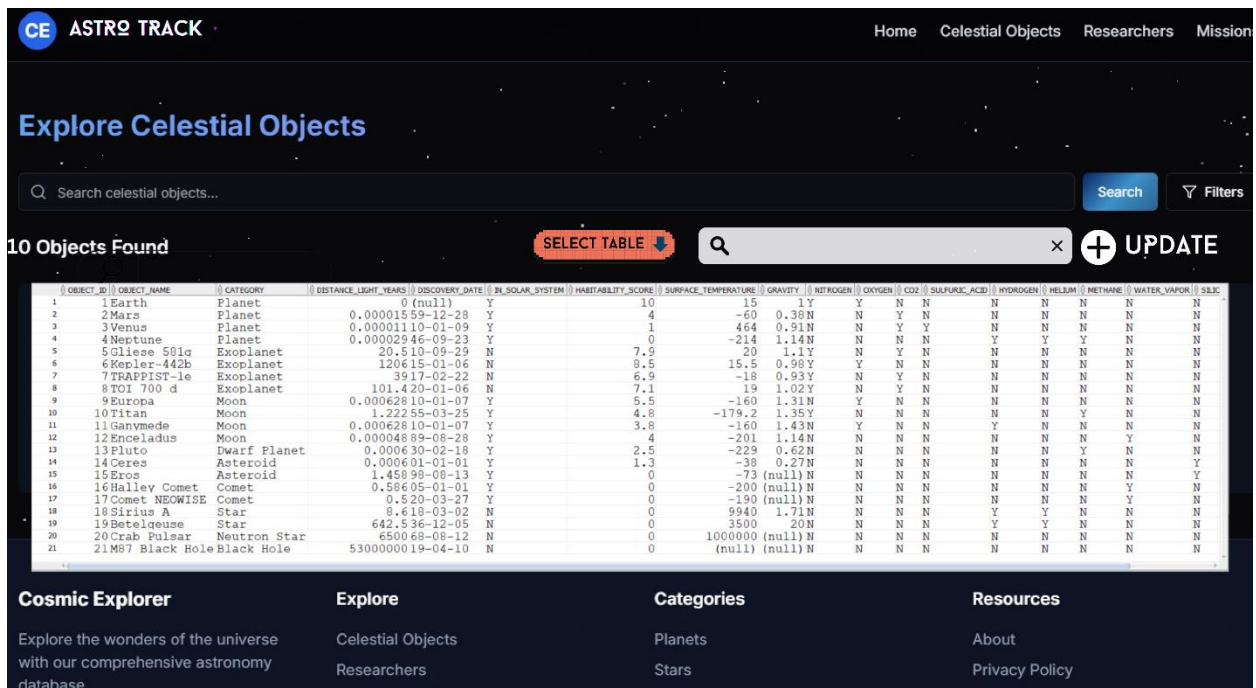
Screenshots of execution of front end with tables content (using programming units)

1. Main page: Astro Track



2. Programming Unit: The process of using PACKAGES of SEQUENCES to INSERT the values into Celestial Objects

2.1. Before inserting values



2.2. Click 'Filter' to display new rows based on your selected criteria. In this case, select the Celestial Objects table.

CE | ASTRO TRACK

Home

Explore Celestial Objects

Search celestial objects...

0 Objects Found

OBJECT_ID	OBJECT_NAME	CATEGORY	DISTANCE_LIGHT_YEARS	DISCOVERY_DATE	IN_SOLAR_SYSTEM	HABITABILITY_SCORE	SURFACE_TEMPERATURE	GRAVITY	NITROGEN	OXYGEN
1	Earth	Planet	0 (null)	Y	10	15	1 Y	Y		
2	Mars	Planet	0.00001559-12-28	Y	4	-60	0.38N	N		
3	Venus	Planet	0.00001110-01-09	Y	1	464	0.91N	N		
4	Neptune	Planet	0.00002946-09-23	Y	0	-214	1.14N	N		
5	511ese 581q	Exoplanet	20.510-09-29	N	7.9	20	1.1 Y	N		
6	Kepler-442b	Exoplanet	120615-01-06	N	8.5	15.5	0.98 Y	Y		
7	TRAPPIST-1e	Exoplanet	3917-02-22	N	6.9	-18	0.93 Y	N		
8	8TOI 700 d	Exoplanet	101.420-01-06	N	7.1	19	1.02 Y	N		
9	Europa	Moon	0.00062810-01-07	Y	5.5	-160	1.31N	Y		
10	Titan	Moon	1.22255-03-25	Y	4.8	-179.2	1.35 Y	N		
11	Ganymede	Moon	0.00062810-01-07	Y	3.6	-160	1.43N	Y		
12	Enceladus	Moon	0.00004889-08-28	Y	4	-201	1.14N	N		
13	Pluto	Dwarf Planet	0.000630-02-18	Y	2.5	-229	0.62N	N		
14	Ceres	Asteroid	0.000601-01-01	Y	1.3	-38	0.27N	N		
15	Eros	Asteroid	1.45898-08-13	Y	0	-73 (null)	N	N		
16	Halley Comet	Comet	0.58605-01-01	Y	0	-200 (null)	N	N		
17	Comet NEOWISE	Comet	0.520-03-27	Y	0	-190 (null)	N	N		
18	Sirius A	Star	8.618-03-02	N	0	9940	1.71N	N		
19	Betelgeuse	Star	642.536-12-05	N	0	3500	20N	N		
20	Crab Pulsar	Neutron Star	650068-08-12	N	0	1000000 (null)	N	N		
21	M87 Black Hole	Black Hole	5300000019-04-10	N	0	(null) (null)	N	N		

Cosmic Explorer

Explore the wonders of the universe with our comprehensive astronomy database.

Explore

Celestial Objects

Researchers

Categories

Planets

Stars

Filter Celestial Objects

Refine your search with these filters

Category

All Categories

Habitability Score (Minimum)

0

Solar System

All Objects

Atmosphere

☐ Has Oxygen

☐ Has Water Vapor

Clear Filters

INSERT

CE | ASTRO TRACK

Home

Explore Celestial Objects

Search celestial objects...

0 Objects Found

OBJECT_ID	OBJECT_NAME	CATEGORY	DISTANCE_LIGHT_YEARS	DISCOVERY_DATE	IN_SOLAR_SYSTEM	HABITABILITY_SCORE	SURFACE_TEMPERATURE	GRAVITY	NITROGEN	OXYGEN
1	Earth	Planet	0 (null)	Y	10	15	1 Y	Y		
2	Mars	Planet	0.00001559-12-28	Y	4	-60	0.38N	N		
3	Venus	Planet	0.00001110-01-09	Y	1	464	0.91N	N		
4	Neptune	Planet	0.00002946-09-23	Y	0	-214	1.14N	N		
5	511ese 581q	Exoplanet	20.510-09-29	N	7.9	20	1.1 Y	N		
6	Kepler-442b	Exoplanet	120615-01-06	N	8.5	15.5	0.98 Y	Y		
7	TRAPPIST-1e	Exoplanet	3917-02-22	N	6.9	-18	0.93 Y	N		
8	8TOI 700 d	Exoplanet	101.420-01-06	N	7.1	19	1.02 Y	N		
9	Europa	Moon	0.00062810-01-07	Y	5.5	-160	1.31N	Y		
10	Titan	Moon	1.22255-03-25	Y	4.8	-179.2	1.35 Y	N		
11	Ganymede	Moon	0.00062810-01-07	Y	3.6	-160	1.43N	Y		
12	Enceladus	Moon	0.00004889-08-28	Y	4	-201	1.14N	N		
13	Pluto	Dwarf Planet	0.000630-02-18	Y	2.5	-229	0.62N	N		
14	Ceres	Asteroid	0.000601-01-01	Y	1.3	-38	0.27N	N		
15	Eros	Asteroid	1.45898-08-13	Y	0	-73 (null)	N	N		
16	Halley Comet	Comet	0.58605-01-01	Y	0	-200 (null)	N	N		
17	Comet NEOWISE	Comet	0.520-03-27	Y	0	-190 (null)	N	N		
18	Sirius A	Star	8.618-03-02	N	0	9940	1.71N	N		
19	Betelgeuse	Star	642.536-12-05	N	0	3500	20N	N		
20	Crab Pulsar	Neutron Star	650068-08-12	N	0	1000000 (null)	N	N		
21	M87 Black Hole	Black Hole	5300000019-04-10	N	0	(null) (null)	N	N		

Cosmic Explorer

Explore the wonders of the universe with our comprehensive astronomy database.

Explore

Celestial Objects

Researchers

Categories

Planets

Stars

Filter Celestial Objects

Refine your search with these filters

Category

All Categories

Planet

Exoplanet

Moon

Star

Galaxy

Gas Giant

Hot Jupiter

Red Supergiant

Clear Filters

INSERT

2.3. Click “Insert” to INSERT the values by using the PACKAGES of SEQUENCES

CE

ASTRO TRACK

Home

Filter Celestial Objects

Refine your search with these filters

Category

All Categories

Habitability Score (Minimum)

Solar System

All Objects

All Objects

In Solar System

Outside Solar System

Clear Filters

INSERT

Explore Celestial Objects

Search celestial objects...

0 Objects Found

OBJECT_ID	OBJECT_NAME	CATEGORY	DISTANCE_LIGHT_YEARS	DISCOVERY_DATE	BULSOLAR_SYSTEM	HABITABILITY_SCORE	SURFACE_TEMPERATURE	GRAVITY	NITROGEN	OXYGEN
1	Earth	Planet	0 (null)	Y	1	15	1 Y	Y		
2	2 Mars	Planet	0.000015 59-12-28	Y	4	-60	0.38 N	N		
3	3 Venus	Planet	0.000011 10-01-09	Y	1	464	0.91 N	N		
4	4 Neptune	Planet	0.000029 46-09-23	Y	0	-214	1.14 N	N		
5	5 Giese 581c	Exoplanet	20.510-09-29	N	7.9	20	1.1 Y	N		
6	6 Kepler-442b	Exoplanet	120615-01-06	N	8.5	15.5	0.98 Y	Y		
7	7 TRAPPIST-1e	Exoplanet	3917-02-22	N	6.9	-18	0.93 Y	N		
8	8 TOI 700 d	Exoplanet	101.420-01-06	N	7.1	19	1.02 Y	N		
9	9 Europa	Moon	0.000628 10-01-07	Y	5.5	-160	1.31 N	Y		
10	10 Titan	Moon	1.222 55-03-25	Y	4.8	-179.2	1.35 Y	N		
11	11 Ganymede	Moon	0.000628 10-01-07	Y	3.8	-160	1.43 N	Y		
12	12 Enceladus	Moon	0.000048 89-08-28	Y	4	-201	1.14 N	N		
13	13 Pluto	Dwarf Planet	0.000630-02-18	Y	2.5	-229	0.62 N	N		
14	14 Ceres	Asteroid	0.000601-01-01	Y	1.3	-38	0.27 N	N		
15	15 Eros	Asteroid	1.458 98-08-13	Y	0	-73 (null)	N	N		
16	16 Halley Comet	Comet	0.586 05-01-01	Y	0	-200 (null)	N	N		
17	17 Comet NEOWISE	Comet	0.520-03-27	Y	0	-190 (null)	N	N		
18	18 Sirius A	Star	8.618-03-02	N	0	9940	1.71 N	N		
19	19 Betelgeuse	Star	642.5 36-12-05	N	0	3500	20 N	N		
20	20 Crab Pulsar	Neutron Star	6500 68-08-12	N	0	1000000 (null)	N	N		
21	21 M87 Black Hole	Black Hole	53000000 19-04-10	N	0	(null)	(null) N	N		

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Explore the wonders of the universe with our comprehensive astronomy database

Explore

Celestial Objects

Researchers

Categories

Planets

Stars

2.4. Click "All Statuses" to view the full list.

CE

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Home

Celestial Objects

Researchers

Missions

Space Missions

Filter Missions

Mission Status

Affiliation

All Statuses

All Affiliations

Clear Filters

✓ All Statuses

Active

Completed

ID	OBJECT_ID	OBJECT_NAME	CATEGORY	DISTANCE_LIGHT_YEARS	DISCOVERY_DATE	PL_SOLAR_SYSTEM	HABITABILITY_SCORE	SURFACE_TEMPERATURE	GRAVITY	NITROGEN	OXYGEN	CO2	SULFURIC_ACID	HYDROGEN	HELIUM	METHANE	WATER_VA
1	1	Earth	Planet	0 (null)	Y		10	15	1Y	Y	N	N		N	N	N	N
2	2	Mars	Planet	0.00001559-12-28	Y		4	-60	0.38N	N	Y	N		N	N	N	N
3	3	Venus	Planet	0.00001110-01-09	Y		1	464	0.91N	N	Y	Y		N	N	N	N
4	4	Neptune	Planet	0.00002946-09-23	Y		0	-214	1.14N	N	N	N		Y	Y	Y	N
5	5	Gliese 581g	Exoplanet	20.510-09-29	N		7.9	20	1.1Y	N	Y	N		N	N	N	N
6	6	Kepler-442b	Exoplanet	120615-01-06	N		8.5	15.5	0.98Y	Y	N	N		N	N	N	N
7	7	TRAPPIST-1e	Exoplanet	3917-02-22	N		6.9	-18	0.93Y	N	Y	N		N	N	N	N
8	8	TOI 700 d	Exoplanet	101.420-01-06	N		7.1	19	1.02Y	N	Y	N		N	N	N	N
9	9	Europa	Moon	0.00062810-01-07	Y		5.5	-160	1.31N	Y	N	N		N	N	N	N
10	10	Titan	Moon	1.22255-03-25	Y		4.8	-179.2	1.35Y	N	N	N		N	N	Y	N
11	11	Ganymede	Moon	0.00062810-01-07	Y		3.8	-160	1.43N	Y	N	N		Y	N	N	N
12	12	Enceladus	Moon	0.00004960-08-28	Y		A	-201	1.14N	N	N	N		N	N	N	Y

2.5. After inserting values result.

CE

ASTRO TRACK

Home

Celestial Objects

Researchers

Missions

Astronomy Researchers

Search Researchers

Search

Q

Search by name or email...

Search

Clear Filters

Affiliation

All Affiliations

0 Researchers Found

ID	OBJECT_ID	OBJECT_NAME	CATEGORY	DISTANCE_LIGHT_YEARS	DISCOVERY_DATE	BL_SOLAR_SYSTEM	HABITABILITY_SCORE	SURFACE_TEMPERATURE	GRAVITY	NITROGEN	OXYGEN	CO2	SULFURIC_ACID	HYDROGEN	HELIUM	METHANE	WATER_VA
12	12	Endeavour	MOON	0.00004859	05-26	Y	4	-201	1.14	N	N	N	N	N	N	N	1
13	13	Pluto	Dwarf Planet	0.000630	02-18	Y	2.5	-229	0.62	N	N	N	N	N	N	Y	N
14	14	Ceres	Asteroid	0.000601	01-01	Y	1.3	-38	0.27	N	N	N	N	N	N	N	N
15	15	Eros	Asteroid	1.45896	08-13	Y	0	-73 (null)	N	N	N	N	N	N	N	N	N
16	16	Halley Comet	Comet	0.58605	01-01	Y	0	-200 (null)	N	N	N	N	N	N	N	N	Y
17	17	Comet NEOWISE	Comet	0.520	03-27	Y	0	-190 (null)	N	N	N	N	N	N	N	N	Y
18	18	Sirius A	Star	9.619	03-02	N	0	9940	1.71	N	N	N	Y	Y	N	N	N
19	19	Betelgeuse	Star	642.536	12-05	N	0	3500	20	N	N	N	Y	Y	N	N	N
20	20	Crab Pulsar	Neutron Star	6500	68-08-12	N	0	1000000 (null)	N	N	N	N	N	N	N	N	N
21	21	M87 Black Hole	Black Hole	53000000	19-04-10	N	0	(null) (null)	N	N	N	N	N	N	N	N	N
22	22	Proxima b	Exoplanet	4.2416	08-24	N	7	0	1.1	Y	N	N	N	N	N	N	N
23	23	OGLE-2016-BLG-1195Lb	Exoplanet	13000	17-04-26	N	6.5	-220	0.75	N	Y	N	N	N	N	N	N